AetherStyle

A Multimodal Content Creation Toolkit for AMD Radeon GPUs / ROCm

Track 1: Development of Multimodal Content Creation Tools

Submission materials — Project Profile Document

1. Project Background

The explosive growth of short-video, social, and e-commerce content has created huge demand for fast, affordable, and private AI-generated images and video. Most open-source AIGC tooling, however, is built and tested for NVIDIA CUDA, leaving AMD Radeon owners — who often have very capable, cost-effective GPUs — underserved.

AetherStyle was created to close that gap. It is a local-first, one-stop multimodal creation toolkit that runs natively on AMD ROCm and turns natural-language prompts into images and short videos. Rather than hard-wiring a single heavy model, AetherStyle observes the machine's disk and VRAM budget and automatically selects the best model that actually fits, while gracefully falling back to lighter models when a download or load fails. The result is a tool that is stable across a wide range of AMD hardware and never crashes on a misconfigured box.

A second design goal is accessibility for Chinese-speaking creators. The video path is built around Wan2.2-TI2V-5B, whose umt5-xxl text encoder understands Chinese natively, so users can write prompts in Chinese without any translation step.

2. Target Users & Application Scenarios

Target users

- Independent content creators and social-media operators (Douyin / Bilibili / YouTube / Instagram).
- E-commerce sellers who need product and marketing visuals on demand.
- Game / film studios and illustrators producing concept art and storyboards.
- Educators who turn lesson points into visuals and short clips.
- Privacy-sensitive teams that require fully local, on-device generation (no cloud upload).
- Hobbyists and small studios running on AMD Radeon hardware.

Application scenarios

- Text-to-video ad spots, vlogs, and explainer clips (720P @ 24fps, a few seconds each).
- Text-to-image product photography, posters, and social cards.
- Image-to-image restyling and concept exploration from an uploaded reference.
- Rapid previsualization for games / films directly on affordable AMD GPUs.
- Local, private generation for commercial or confidential material.

3. System Architecture

AetherStyle is organized in clear layers so that model choice, hardware adaptation, and the user interface are decoupled:

- UI layer — a Gradio Web UI (port 7860) with dedicated tabs for Text-to-Image, Image-to-Image, and Text-to-Video, plus a model-management panel. An optional FastAPI service exposes the same engine programmatically.

- Orchestration layer (engine.py) — routes each request to the best-fit pipeline, enforces disk/VRAM gating, loads and caches models, runs the tiered fallback chain, normalizes outputs, and exports final files.
- Model layer — diffusers pipelines: SD3.5-Medium / SDXL / SD v1.5 for images; Wan2.2-TI2V-5B / LTX-Video / Legacy damo-vilab SDPipeline for video.
- Runtime layer — PyTorch on the ROCm (HIP) backend, bf16 compute, with CPU offload when VRAM is tight.

Data flow: prompt (and optional image) -> engine selects a model that fits the budget -> generation -> frame/image normalization -> export to MP4 / PNG (and optional GIF). Robustness comes from three mechanisms: (a) disk gating that refuses to download weights that will not fit; (b) weight-integrity checks that detect incomplete downloads; and (c) an automatic tier fallback that routes to the next model when any stage fails — so the service stays up instead of crashing.

4. Model & Algorithm Introduction

Image generation

Image models are latent-diffusion / DiT architectures. The primary model is Stable Diffusion 3.5-Medium (MMDiT with a T5-XXL + CLIP text encoder and flow-matching sampling), chosen for its strong prompt adherence and quality. It falls back to SDXL (UNet + refiner) and then SD v1.5 for constrained environments.

Video generation

The primary video model is Wan2.2-TI2V-5B: a ~5B-parameter DiT with the umt5-xxl multilingual text encoder, capable of 720P @ 24fps clips under the Apache-2.0 license. Its multilingual encoder means Chinese prompts work natively. Two fallbacks keep the tool usable everywhere: LTX-Video (a 2B DiT with T5-XXL, very fast) and the legacy damo-vilab text-to-video-ms-1.7b pipeline.

Inference configuration

- Precision: bf16 by default (ROCm-friendly); overridable via AETHER_T2V_DTYPE.
- Sampling: 50 steps and guidance scale 6.0 for quality; classifier-free guidance with optional negative prompts.
- Constraints handled automatically: resolution divisible by 16; (num_frames - 1) divisible by 4.
- Reproducibility: optional fixed seed for deterministic outputs.

The key algorithmic contribution at the system level is the best-model selection heuristic: it compares each candidate model's size against free disk and VRAM, downloads only what fits, and degrades safely — turning 'will it run on my AMD card?' into an automatic decision.

5. Adaptation for AMD Radeon GPU / ROCm

- Runs on the ROCm (HIP) backend via the official PyTorch ROCm wheels (rocm6.0 / rocm7.0 index URLs); no CUDA-specific code is used.
- Validated on an AMD Radeon GPU (gfx1100) with 48 GB VRAM, single-card, driving both image and video generation.
- Uses bf16 compute and CPU offload so large models (Wan 34 GB, SD3.5 10 GB) coexist on a single consumer/workstation AMD card without OOM.
- Portable across the AMD stack: because the code relies only on diffusers + torch.cuda (which ROCm maps to HIP), the same binaries run on other Radeon gfx targets.

- Compatibility shims for older toolchains: e.g. HSA_OVERRIDE_GFX_VERSION=11.0.0 when an older torch build lacks a gfx1100 kernel; a HfFolder stub so Gradio loads on newer huggingface_hub.
- Disk/VRAM gating and tiered fallback are specifically engineered so the tool remains stable on constrained AMD boxes instead of failing mid-run.

In summary, AetherStyle makes multimodal image+video generation a first-class, stable experience on AMD Radeon / ROCm hardware, with automatic model selection, native Chinese prompts, and crash-proof operation.